

ECE 501 DIGITAL IMAGE PROCESSING

Weekly Progress Report - Week 1

Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

GROUP 10

Team Members

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Repository Link:

https://github.com/samridhim18/DIP_2026_ProjectNo_10

Report Submission Date:

12th September 2026

Work Done This Week

- Evaluated the ECE 501 project guidelines and defined the project scope: Segmentation and Quantitative Analysis of Brain MRI Tumor Using Classical Image Processing Techniques.
- Selected the BraTS2020 dataset (in NIfTI file format, comprising of T1, T1Gd, T2, and T2-FLAIR volumes and their segmentation masks) as the primary dataset of the project, along with the BraTS 2015 2D-slice dataset as an alternate dataset with reduced complexity.
- Configured the GitHub repository (DIP_2026_Project_10) with appropriate directory structures: Mid_Sem_Report, End_Sem_Report, Results, and Codes.
- Assigned tasks to each member of the group to deal with data acquisition, processing, literature survey, and documentation.
- Conducted an initial literature survey of classical (non-deep learning) MRI segmentation methods emphasizing thresholding, morphological operations, and clustering techniques.

Work To Be Done Next Week

- Download and arrange the BraTS data set. Create scripts to load and visualize MRI slices from samples.
- Do basic pre-processing on the sample images: skull stripping, denoising, and enhancement of the image contrast.
- Segmentation using thresholding will be done as a base technique to benchmark the advanced techniques.
- Add further readings of 2 to 3 papers on morphological and clustering segmentation.
- Write the Introduction and Methodology sections of the report in preparation of the Mid-Semester report.
- Understand few important topics and concepts related to the project and few terms in papers.

Literature Review

- For skull stripping and tumor separation, there is a paper which proposes hybrid multilevel thresholding technique on brain MRIs, using threshold segmentation combined with morphological post-processing for obtaining tumor segment. It can serve as an example of classical application of thresholding and morphology.

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10000536/>)

- Another paper addresses the issues of noise and non-brain tissues first, conducting adaptive Wiener filtering to reduce noise in the image and performing morphological operations to remove all non-brain tissue areas from the image, followed by clustering of the resulting region. Once again, morphological post-processing takes place at the end of the process for further cleaning of the segmented area. This paper provides a nice example of the pipeline structure (denoise - strip - cluster - refine).

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6481128/>)

- However, the work that most resembles our research is the comparison between two classical algorithms that do not involve machine learning, wherein one involves the use of simple morphological operations while the other uses the discrete wavelet transform combined with simple morphological processing. These two algorithms share a common first step of skull stripping by binary erosion and then contrast enhancement and histogram thresholding.

(<https://www.jyi.org/2017-august/2017/8/19/brain-tumor-segmentation-using-morphological-processing-and-the-discrete-wavelet-transform>)

- Finally, a comparative study compares five classical methods against each other - region growing, thresholding, watershed, fuzzy C-means, and histogram thresholding, all of which have been performed on skulls after their morphological segmentation. It would be useful for explaining the reasons behind the choice of a particular classical method because they are compared based on accuracy and speed of the processing. (Note: The entire text is behind the paywall; we use an abstract here only.)

(<https://www.sciencedirect.com/science/article/abs/pii/S2214785321028285>)

Challenges Faced

- Limited time available in Week 1 because of late information regarding the deadline for submitting the assignment, limiting the scope of dataset exploration and literature survey that could be performed this week.
- The BraTS dataset consists of big 3D volumes in the format of NIfTI files, thus, extra preprocessing steps would be needed to transform them into an appropriate format for classical 2D image processing methods.
- Selecting only one classical method (thresholding, clustering, or morphology), as multiple approaches have been proposed in literature.